CSC2106: IoT Protocols and Networks
Project Proposal Report

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Project Title
LoRa Mesh network for Deep Tunnel Infrastructure Connectivity in Singapore

1. What specific issue or problem has been identified that requires a solution?
Singapore is currently undertaking large-scale deep-tunnel infrastructure projects, such as the Cross Island Line (CRL), with tunnels reaching depths of up to 70 meters, equivalent to a 25-storey HDB block. (Justin O, 2019)

A research study done by IEEE showed that ensuring reliable mobile connectivity in these deep tunnels is challenging due to their closed environments and reinforced concrete structures, which significantly attenuate radio signals through reflection, absorption, and lossy wall effects (Abdus S et al., 2023). Industry analyses of modern building materials further confirmed that rebar-reinforced concrete acts as a partial Faraday cage. Steel reinforcement grids reflect and absorb electromagnetic waves, causing signal losses of up to 30dB or more, equivalent to a 99.9% reduction in power for standard walls, with thicker tunnel walls causing significantly higher attenuation. (Resolution Wireless, 2025)

Current implementations rely on 4G and 5G signals distributed through repeaters, distributed antenna systems (DAS), and leaky feeder cables. However, technical comparisons indicate that leaky feeder systems suffer significant longitudinal signal loss, particularly at frequencies above 1 GHz, which are critical for modern 4G and 5G communications. Due to their linear, single polarized nature, leaky feeders offer no meaningful MIMO or spatial diversity benefits, limiting their ability to mitigate fading. As a result, achievable bandwidth is reduced and latency increased compared to antenna-based solutions. Furthermore, the reliance on multiple in-line amplifiers introduces faults that are difficult to detect, increasing system complexity and operational burden. (Poynting Antennas, 2023)

Additionally, measurement-based studies of 4G and 5G deployments show that while 5G particularly in mid and high-frequency bands can deliver very high throughput, its performance is significantly more sensitive to obstruction, distance, and penetration loss than

lower-frequency systems. Millimeter-wave 5G, in particular, suffers rapid degradation under blockage such as building materials and human body interference, necessitating dense small-cell deployments to maintain performance. These propagation characteristics limit the reliability of high-frequency 5G in complex or obstructed environments compared to lower-band deployments. (Muhammad Iqbal R et al., 2023)

Another critical safety challenge lies in maintaining reliable wireless connectivity for environmental monitoring sensors, such as toxic gas and temperature detectors, without relying on extensive wired or leaky-feeder cabling infrastructure. In tunnel environments, such cabling systems are costly to install and difficult to modify once deployed, particularly as they are often located in hard-to-access areas. Moreover, exposure to high humidity, extreme temperatures, mechanical stress, and corrosion makes these systems vulnerable to degradation and faults that are difficult to detect and repair, increasing operational risk for safety-critical monitoring applications. (Peter B, 2024)

Radio propagation studies in curved tunnels also shows that bends introduce additional attenuation beyond straight-tunnel propagation, with electromagnetic energy tending to concentrate along the concave wall, a phenomenon often described using whispering-gallery-mode analogies. Non-line-of-sight propagation around tunnel bends experiences significant extra loss due to the rapid decay of higher-order modes, with the severity of attenuation increasing at higher frequencies. As frequency increases into the GHz range, signals become increasingly sensitive to curvature and obstruction, meaning that without intermediate repeaters, distributed antennas, or multi-hop mesh topologies, tunnel bends can create localized coverage gaps, leaving workers and environmental sensors situated beyond the bend in radio dead zones with no reliable connectivity. (Bo A et al., 2015)

Collectively, these findings indicate that conventional high-frequency cellular and cable-based systems are fundamentally mismatched to deep-tunnel environments, motivating the exploration of low-frequency, multi-hop wireless alternatives capable of adapting to tunnel geometry and structural attenuation.

2. Why is your topic important or relevant to industry, academia, or society?

This topic addresses critical infrastructure safety and operational efficiency challenges in Singapore's ongoing and future underground construction projects. Large-scale deep-tunnel systems, such as the Deep Tunnel Sewerage System (DTSS), represent long-term national infrastructure requiring high levels of reliability and operational assurance. DTSS Phase 2 involves the construction of a 98 km-long network of deep tunnels and link sewers serving the western half of Singapore, underscoring the scale and complexity of underground works and the need for robust monitoring of tunnelling activities and environmental conditions.
(PUB, 2025; Justin O, 2019)

Industrial relevance: The construction and infrastructure sector increasingly requires cost-effective alternatives to traditional wired communication systems. Studies of underground communication infrastructure indicate that leaky feeder and other wired systems are dominated

by high cable, installation, and maintenance costs and are difficult to reconfigure once deployed. Industry analyses report that installed wired instrumentation can cost approximately USD $10–20 per foot, with even short conduit runs of 50–75 feet costing up to USD $1,500, while wireless monitoring systems can reduce on-site installation labor by 50–75% by eliminating trenching and conduit works and enabling flexible, scalable expansion as construction requirements evolve. (Dana T, 1980; BBP, n.d.)

Academic relevance: Prior research in underground mining and tunnel communications has identified multi-hop LoRa mesh networks as a promising solution for extending coverage in subterranean environments where conventional LoRaWAN deployments are impractical. However, existing studies are largely limited to mining or linear tunnel scenarios and rely predominantly on simulation or simplified flooding-based approaches, with limited comparative analysis of dense LoRa mesh performance in complex civil infrastructure settings. This project addresses this gap by investigating LoRa mesh performance in vertically constrained and structurally obstructed underground environments, contributing new insights relevant to urban deep-tunnel infrastructure contexts. (Nalith U et al., 2025)

Societal Impact: Under Singapore's MOM Workplace Safety and Health (WSH) Regulations, real-time gas monitoring is mandatory for confined and underground work environments where toxic or flammable gases may be present. In deep-tunnel infrastructure projects such as MRT and DTSS, deploying continuous monitoring is challenging due to vertical depth and structural obstructions. (Minerva, n.d.) A LoRa mesh–based wireless sensor network enables MOM-compliant, real-time gas detection without reliance on fixed cabling, maintaining connectivity around tunnel bends and shafts. This improves emergency response coordination and significantly enhances worker safety in Singapore's underground construction environments.

3. What is the potential impact of addressing this challenge?

The successful implementation of a LoRa-based mesh or multi-hop wireless network within Singapore's deep-tunnel infrastructure represents a significant shift away from rigid, cable-dependent communication systems towards a more resilient, flexible, and cost-efficient connectivity model. By directly addressing the propagation, installation, and maintenance challenges inherent in subterranean environments, this approach enables a communication architecture that is better suited to the complex geometry and harsh conditions of deep underground infrastructure.

Economic Impact and Infrastructure Cost Efficiency: One of the most significant impacts of adopting a LoRa-based wireless sensor network is the reduction in installation and lifecycle costs associated with underground monitoring systems. Conventional wired solutions rely heavily on extensive cabling, conduits, and repeaters, which incur high material, labour, and maintenance costs that scale linearly with tunnel length (MDPI, 2024). In large-scale tunnel projects spanning several kilometres, such costs quickly become prohibitive.

By eliminating the need for leaky feeders and extensive hardwired connections, a wireless LoRa-based architecture significantly reduces both upfront installation expenses and long-term maintenance requirements. Owing to its long-range, low-frequency propagation characteristics, LoRa outperforms WPAN and WLAN technologies such as ZigBee and Wi-Fi in underground environments, requiring fewer gateways or relay nodes to achieve comparable coverage (IEEE, 2024; MDPI, 2024). This reduction in infrastructure complexity directly translates into improved cost efficiency and lower operational expenditure over the lifespan of the tunnel system.

Operational Agility and Scalability: Addressing underground connectivity challenges using a LoRa-based mesh architecture introduces operational flexibility that is difficult to achieve with traditional linear systems such as leaky feeders. LoRa-based wireless sensor networks are inherently scalable, allowing additional sensor nodes or relay devices to be deployed incrementally as tunnel excavation progresses, without requiring major modifications to existing infrastructure (MDPI, 2024).

This flexibility enables a “plug-and-play” deployment model, where sensors can be repositioned or added to accommodate evolving construction phases and monitoring requirements. Although LoRa trades off data rate to achieve long-range, low-power communication, this characteristic is well suited for environmental monitoring applications, where periodic measurements must balance responsiveness with battery longevity (ScienceDirect, 2023). As a result, the system supports continuous monitoring while minimising maintenance effort and power consumption.

Mission-Critical Safety and Disaster Resilience: The most critical impact of this project lies in its contribution to workplace safety and regulatory compliance. By extending reliable connectivity into areas previously affected by signal dead zones caused by tunnel curvature, reinforced concrete, or depth, the proposed network enables real-time monitoring of toxic gases and environmental conditions in accordance with Singapore’s Workplace Safety and Health (WSH) regulations.

Unlike wired systems, which are vulnerable to failure due to humidity, corrosion, mechanical damage, or explosions, a wireless mesh-based architecture offers inherent redundancy and self-healing capabilities. Through multi-hop communication, sensor data can be dynamically rerouted if individual nodes are damaged or obstructed, maintaining connectivity between underground sensors and surface control rooms (IEEE, 2024; MDPI, 2024). This resilience enhances early warning capabilities for hazardous conditions and supports predictive maintenance strategies, ultimately improving worker safety and reducing operational risk in deep-tunnel environments.

4. What specific gap or challenge in the field does this work address?

This project addresses the critical gap between LoRa technology's theoretical capabilities and its practical implementation in Singapore's unique deep-tunnel infrastructure context.

Technical Gap: Existing research on LoRa-based mesh networks predominantly evaluates performance in horizontally distributed or radially expanded network topologies, with routing and

propagation models primarily based on distance to a gateway in planar urban environments. Recent comprehensive studies, such as Durand and Booysen (2025), rely on large-scale simulations that model path loss using log-distance attenuation and assume static, surface-level deployments, without consideration of vertical separation, reinforced concrete barriers, or deep underground geometries.

In contrast, Singapore's underground infrastructure such as the Deep Tunnel Sewerage System (DTSS Phase 2) and MRT stations feature deep vertical shafts, multi-level structures, and thick reinforced concrete linings, which introduce substantially different radio propagation conditions. While these structural characteristics are well documented in civil engineering literature, empirical evaluation of LoRa mesh performance in vertically obstructed underground environments remains limited (Stellar Structures, 2025). This gap motivates targeted investigation of LoRa mesh propagation in deep, reinforced urban underground infrastructure.

Implementation Gap: Conventional LoRaWAN deployments rely on a gateway-centric, single-hop star topology in which end devices must maintain a sufficiently reliable direct radio link to a gateway. While this architecture performs well in open or moderately obstructed environments, it becomes increasingly fragile in subterranean contexts such as deep tunnels, shafts, and underground infrastructure, where reinforced concrete, depth, and complex geometry significantly attenuate radio signals and where gateway placement is highly constrained. Existing literature highlights that such environments often suffer from insufficient direct coverage rather than network congestion, resulting in isolated end devices that are unable to reliably communicate with gateways.

To address coverage limitations, multi-hop and mesh-based LoRa networks have been actively explored in research, demonstrating improvements in packet delivery ratio for nodes located far from centralized gateways. However, despite this growing body of work, there remains no standardized multi-hop LoRa protocol. Current standardization efforts, such as the LoRaWAN Relay specification, provide only a limited single-hop extension intended to bridge short communication gaps and explicitly do not support relay chaining or full mesh routing. (thethingsnetwork, n.d.)

Consequently, relay-based solutions cannot form self-healing or multi-tier networks suitable for vertically segmented underground infrastructure. Furthermore, although relay support is designed to interoperate across LoRaWAN versions, its operation requires additional protocol capabilities beyond the base LoRaWAN Layer-2 specification, implying that legacy end devices may require firmware or stack upgrades to participate (LoRa Alliance, 2024). This combination of architectural constraints and standardization gaps highlights an unresolved challenge in integrating mesh-style LoRa networks with existing LoRaWAN infrastructure, particularly for deep underground environments where multi-hop vertical connectivity and backward-aware gateway integration are critical for worker safety monitoring and infrastructure sensing.

Protocol Challenge: The LoRaWAN Relay specification (TS011) introduces a standardized mechanism for extending coverage in scenarios where end devices cannot reliably

communicate directly with a gateway due to distance or obstructions. However, this mechanism is explicitly designed as a single-hop relay extension rather than a general-purpose repeater or mesh routing protocol. According to TS011, end devices intending to operate via a relay are required to implement LoRaWAN Layer-2 version 1.0.4 or later, along with relay-specific features such as Wake-on-Radio (WOR) signalling, Channel Activity Detection (CAD) behaviour, and relay-specific security key derivation. (LoRa Alliance, 2024) These requirements imply that legacy LoRaWAN end devices, particularly those conforming to earlier versions such as LoRaWAN 1.0.2 and below cannot participate in relay-assisted communication without firmware updates or protocol stack modifications.

Furthermore, the relay mechanism defined in TS011 does not operate as a transparent packet forwarder. Instead, the relay terminates the original end-device transmission, encapsulates the payload with additional relay metadata, and generates a new LoRaWAN uplink frame addressed to the network server. From the network's perspective, the relay itself appears as a standard LoRaWAN end device operating on a reserved application port. This design choice enables compatibility with existing gateways and network servers that support the relay specification, but it also constrains relay operation to a single-hop, non-chained topology. TS011 explicitly does not support relay chaining, multi-hop routing, or autonomous mesh formation, and therefore cannot provide self-healing or multi-tier connectivity in complex underground or vertically segmented environments. (thethingsnetwork, n.d.)

As a consequence of these architectural constraints, the standardized LoRaWAN relay mechanism primarily addresses localized coverage gaps rather than structural connectivity challenges. In deployment scenarios involving deep tunnels, shafts, or heavily reinforced concrete infrastructure where signal attenuation may isolate entire groups of sensors rather than individual nodes, as such the relay specification offers limited applicability. In addition, because relay participation requires explicit protocol support at the end-device level, heterogeneous networks containing legacy LoRaWAN devices cannot be seamlessly integrated into relay-based extensions.

Routing Algorithm Gap: Existing research on subterranean LoRa mesh networks has primarily focused on extending coverage through flooding-based or broadcast-oriented forwarding mechanisms, particularly in linear underground environments such as tunnels, mines, and pipelines. While these approaches offer simplicity and robustness, they incur excessive retransmissions, and limited scalability, making them unsuitable for dense or long-term deployments. Moreover, many current solutions remain custom-built and application-specific, with no standardized routing framework to support autonomous self-discovery, self-healing, and efficient multi-hop operation in large-scale underground networks. (Nalith U et al., 2025; Guus L et al., 2023)

Recent studies have introduced position-aware routing strategies to mitigate flooding overhead and improve efficiency in subterranean LoRa mesh networks. However, these solutions are largely designed and evaluated for linear or planar underground topologies, and do not explicitly address the challenges posed by complex, vertically segmented underground infrastructures. In deep tunnel systems such as Singapore's transport, utility, and service tunnels nodes must

maintain reliable connectivity across long horizontal stretches, vertical shafts, and multiple depth levels. This creates unresolved challenges in scalable route establishment without continuous flooding, intelligent gateway nomination, and robust route recovery under link failures. This work addresses these gaps by targeting adaptive LoRa mesh networking tailored for deep tunnel infrastructure connectivity, extending existing subterranean LoRa research toward realistic, large-scale underground deployments.

5. What are the main goals of this project or research?

Implementation of a Version-Blind Abstraction Layer
The first goal of this project is to implement a pseudo-relay mechanism that allows LoRaWAN packets to be forwarded across multiple hops in an underground environment. The system uses a simple encapsulation approach where the LoRa PHYPayload is treated as raw data and forwarded without being decoded or modified.

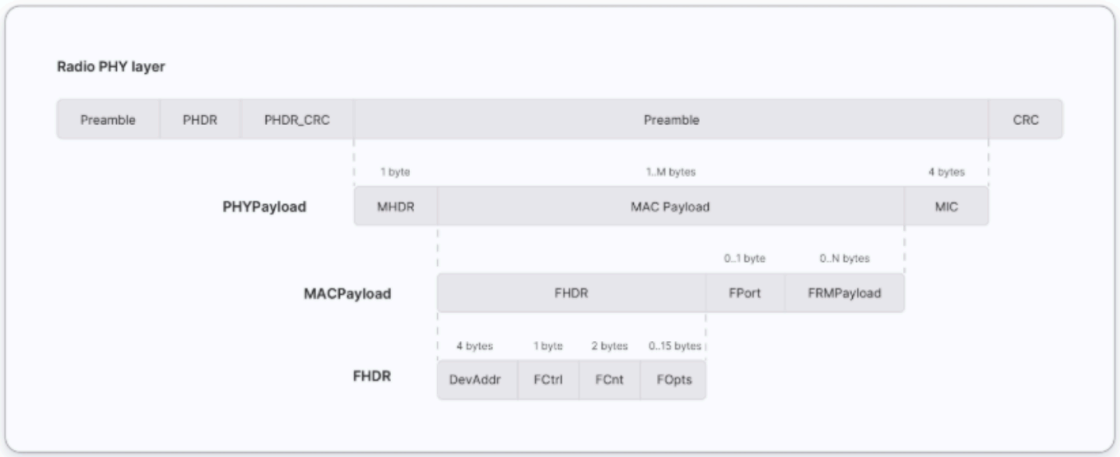
By doing this, the mesh network operates independently of the LoRaWAN protocol version used by the end devices (e.g. v1.0.2 or v1.0.4). This allows existing and legacy LoRaWAN sensor nodes to participate in the multi-hop network without requiring firmware changes, while keeping the back-end system unaware of the internal mesh structure.

Custom Frame Image

Data Messages

There are 4 data message types used in both LoRaWAN 1.0.x and 1.1. These data message types are used to transport both MAC commands and application data which can be combined together in a single message. Data messages can be confirmed or unconfirmed. Confirmed data messages must be acknowledged by the receiver whereas unconfirmed data messages do not need to be acknowledged by the receiver.

A data message is constructed as shown below:



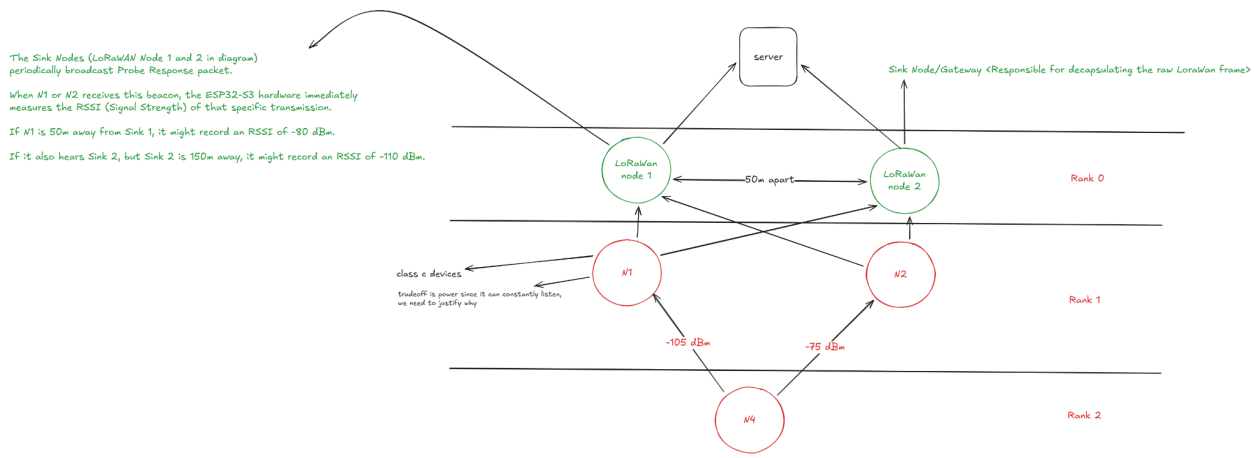
Heuristic Path Optimization and Multi-Sink Intelligence

The project aims to implement a simple rank-based routing approach to improve reliability and efficiency in tunnel environments. Each node selects its parent node by computing a weighted score based on:

- Rank, which represents the number of hops to a gateway, and
- RSSI, which reflects the quality of the wireless link.

This approach allows the network to:

- **Reduce overall Time-on-Air (ToA)** by choosing better-quality links and avoiding unnecessary retransmissions in noisy underground conditions.
- **Support multiple gateways (multi-sink)** by allowing nodes to forward data toward different LoRaWAN gateways. If the connection to one gateway becomes weak or blocked, the node can route its data through an alternative path, improving network reliability.



High-Integrity Deterministic Telemetry Management

A key objective of this project is to support reliable, uplink-dominant communication, in which the majority of network traffic consists of sensor telemetry transmitted from underground nodes to surface gateways. Approximately 90–95% of network traffic is expected to be uplink, carrying safety-critical information such as gas concentration and temperature readings. This traffic pattern reflects the operational requirements of underground monitoring systems, where timely delivery of sensor data is prioritised over bidirectional interaction.

To improve delivery reliability in the presence of fluctuating signal conditions caused by tunnel geometry, reinforced concrete, and interference, the system implements hop-by-hop acknowledgements with bounded retransmissions between mesh nodes. By performing acknowledgements locally at each hop rather than end-to-end, the network can quickly detect link failures and recover without incurring excessive latency. Retransmissions are deliberately limited to prevent congestion, excessive airtime consumption, and cascading collisions, while still providing multiple delivery opportunities for high-priority telemetry.

Downlink communication is intentionally minimised and reserved exclusively for essential control operations, such as configuration updates or parameter adjustments. This design choice reflects the inherent constraints of LoRa-based networks, where downlink transmissions must be tightly scheduled within limited receive windows and consume shared gateway airtime, increasing collision risk in dense deployments. By avoiding frequent downlinks and maintaining an uplink-centric traffic model, the system improves scalability, predictability, and reliability for safety-critical underground sensing applications.

Self-Healing Resilience via Self-Nomination Logic
The project also aims to demonstrate basic self-healing behaviour in the mesh network. When a node detects that its current parent is no longer reachable, it can switch to an alternative parent or forwarding path without manual intervention.

To avoid excessive retransmissions and network congestion, the system applies:
- Hop limits (TTL) to prevent packets from circulating indefinitely, and
- Packet deduplication to discard repeated frames.

These mechanisms help maintain stable operation and improve reliability in dense underground tunnel deployments.

6. What specific outcomes are expected by the end of this study?
By the end of this study, the project is expected to demonstrate the feasibility of LoRa mesh networking for deep underground infrastructure connectivity in Singapore. The project will demonstrate the following:

1. Functional LoRa mesh prototype system
2. Validated multi-hop communication performance
3. Demonstration of reliable coverage extension
4. Resilience and self-healing capabilities
5. Cost and deployment feasibility

Collectively, these outcomes will demonstrate that a LoRa-based mesh network is technically viable, cost-effective and resilient for deep tunnel infrastructure, providing a practical alternative to cable dependent systems while improving safety, flexibility, and maintainability in Singapore's underground tunnel projects.

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